Multiphysical Topology Optimization of an Electric Machine using Topological Derivatives

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CREATOR
COMPUTATIONAL ELECTRIC MACHINE LABORATORY

Introduction

ÖAW RICAM

$$\mathrm{d}\mathcal{J}(\Omega)(z) = \frac{1}{T}\int_0^T \frac{1}{|\omega|}\big(h_\omega(\operatorname{curl}_\xi K_{\operatorname{curl} u(t,z)} + \operatorname{curl} u(t,z)) - h_\omega(\operatorname{curl} u(t,z))$$
$$- h'_\omega(\operatorname{curl} u(t,z))\operatorname{curl}_\xi K_{\operatorname{curl} u(t,z)}, \operatorname{curl} p(t,z)\big)_{\mathbb{R}^2}$$
$$+ \frac{1}{|\omega|}\big((h'_{\Omega^c}(\operatorname{curl} u(t,z)) - h'_\Omega(\operatorname{curl} u(t,z))\operatorname{curl}_\xi K_{\operatorname{curl} u(t,z)}, \operatorname{curl} p(t,z)\big)_\omega$$
$$+ (h_{\Omega^c}(\operatorname{curl} u(t,z)) - h_\Omega(\operatorname{curl} u(t,z))), \operatorname{curl} p(t,z))_2$$
$$+ (\sigma_{\Omega^c} - \sigma_\Omega)\partial_t u_0(t,z), p_0(t,z)\ \mathrm{d}t$$
$$+ 2\lambda_\Omega \frac{(\lambda_{\Omega^c} - \lambda_\Omega)^2}{(\lambda_{\Omega^c} - \lambda_\Omega)}(\nabla_x \vartheta_0(z), \nabla_x \phi(z))_2 - (P_{\Omega^c}(u_0(z))$$
$$- P_\Omega(u_0(z)), \phi(z))_2 + (c_{\Omega^c} - c_\Omega)\vartheta_{\vartheta *}(\vartheta_0(z))$$
$$(h_\omega(\operatorname{curl} K_U + U) - h_\omega(U), \operatorname{curl} v)_{\mathbb{R}^2} = -(h_{\Omega^c}(U) - h_\Omega(U), \operatorname{curl} v)_\omega, \quad \forall v \in BL$$

Introduction

$$
\begin{aligned}
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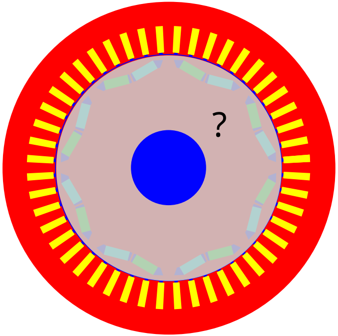
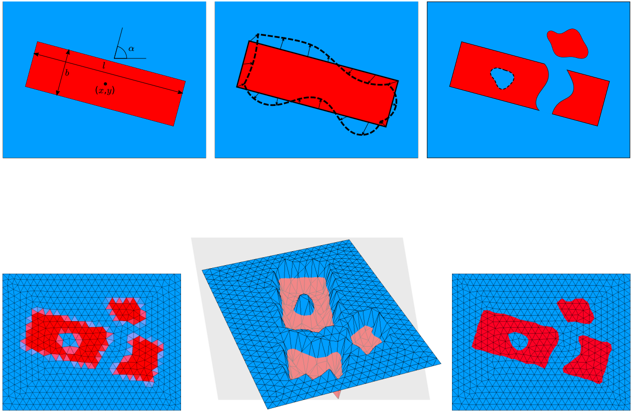
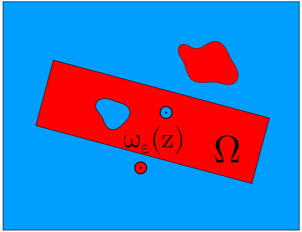
Motivation



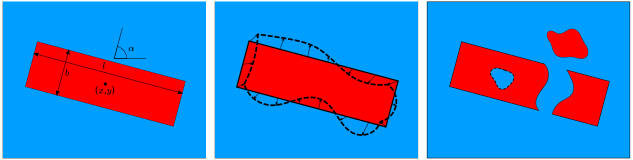


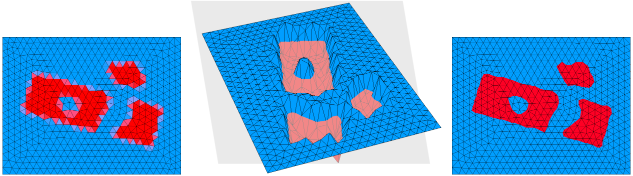
$\rightarrow$ Gradient-based Topology opt. with Level-set representation and Topological derivatives

Motivation



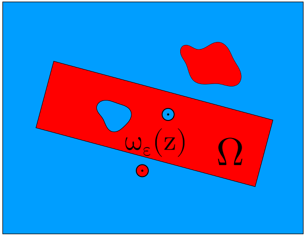
$$\mathrm{d}\mathcal{J}(\Omega)(z) = \lim_{\varepsilon \to 0} \frac{\mathcal{J}(\Omega_\varepsilon(z)) - \mathcal{J}(\Omega)}{|\omega_\varepsilon|}$$





$\rightarrow$ Gradient-based Topology opt. with Level-set representation and Topological derivatives

Finding a minimizer



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Optimality condition

We call a design Ω optimal if

$$\mathrm{d}\mathcal{J}(\Omega)(z) > 0$$

Level set function $\psi \in C(\overline{D})$: $\begin{cases} \psi(x) < 0 & \Leftrightarrow \quad x \in \Omega, \\ \psi(x) > 0 & \Leftrightarrow \quad x \in \Omega^c. \end{cases}$

Lemma

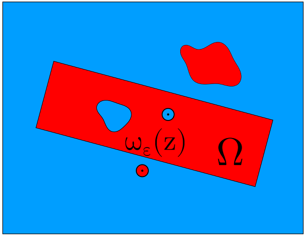
Ω is a local minimizer if for the generalized TD g_Ω

$$c\psi = g_\Omega := \begin{cases} -\mathrm{d}\mathcal{J}(\Omega)(x) & \text{if } x \in \Omega, \\ \mathrm{d}\mathcal{J}(\Omega)(x) & \text{if } x \in \Omega^c \end{cases}, \quad c > 0.$$

Fixed point iteration [Amstutz, Andrä, 2006]

- Compute $\mathcal{J}(\Omega), g_\Omega$
- Update $\psi^{k+1} = (1-s)\psi^k + s g_\Omega$
- Stop if $\angle_{L_2}(\psi, g_\Omega) < \varepsilon$

Finding a minimizer



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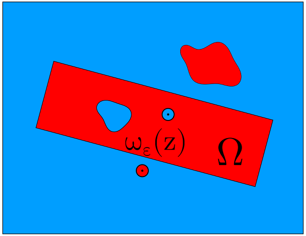

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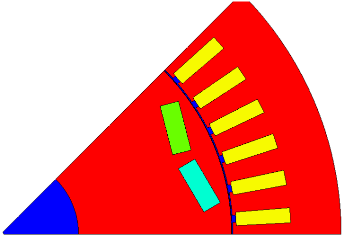
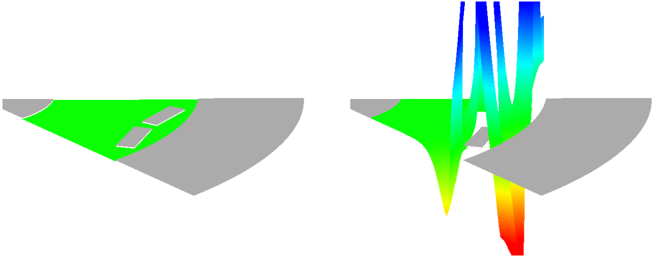
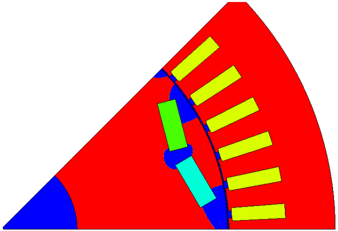
Visualization



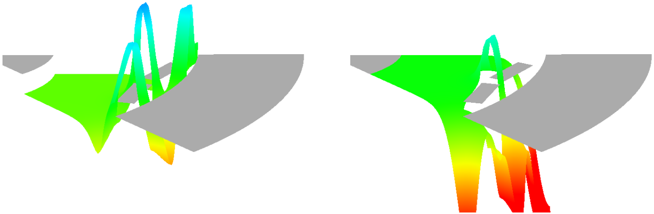
$\mathcal{J}(\Omega) = -569\text{Nm}$



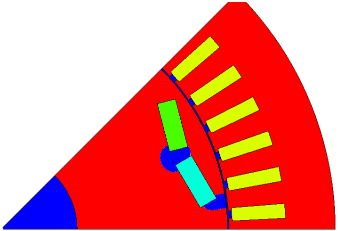
Visualization



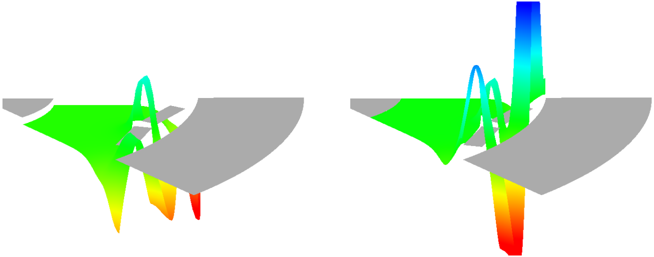
$$\mathcal{J}(\Omega) = -738\text{Nm}$$



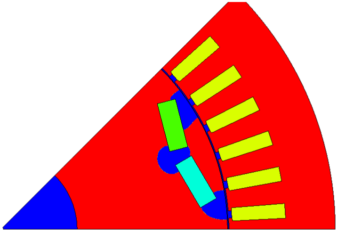
Visualization



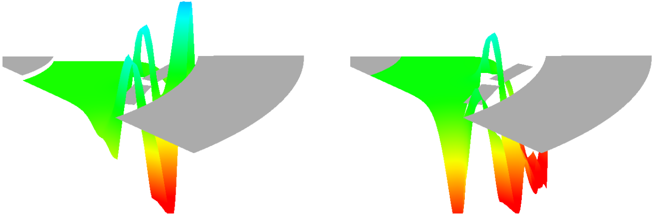
$$\mathcal{J}(\Omega) = -772\mathrm{Nm}$$



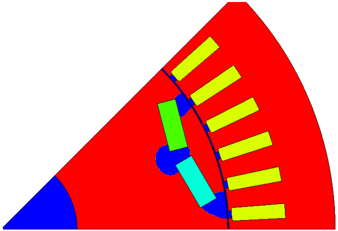
Visualization



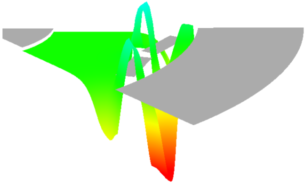
$$\mathcal{J}(\Omega) = -793\text{Nm}$$

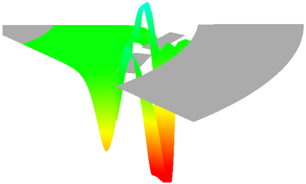


Visualization

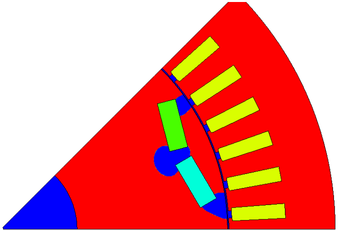


$$\mathcal{J}(\Omega) = -823\text{Nm}$$





Visualization



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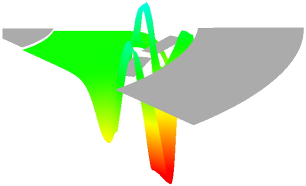
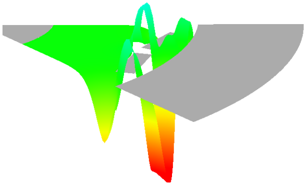
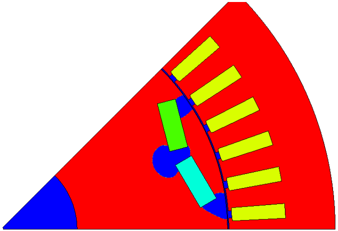
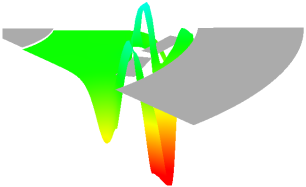


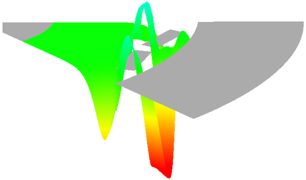


Visualization

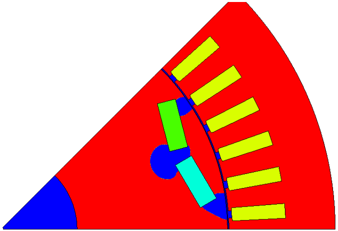


$$\mathcal{J}(\Omega) = -831\mathrm{Nm}$$

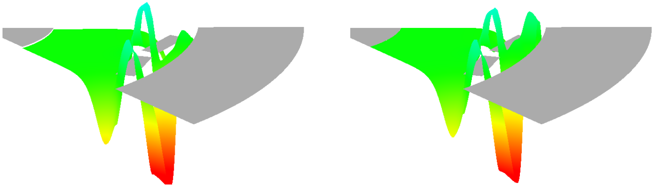




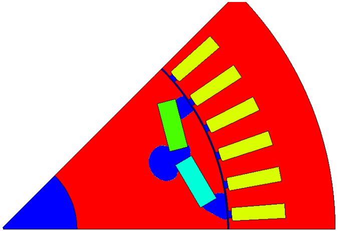
Visualization



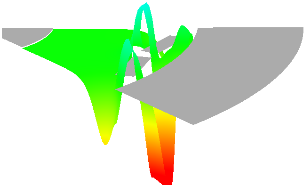
$$\mathcal{J}(\Omega) = -833\text{Nm}$$

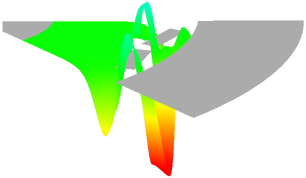


Visualization

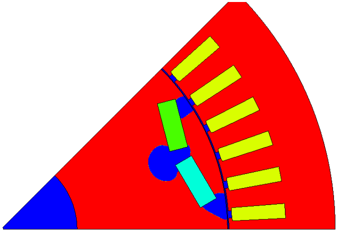


$$\mathcal{J}(\Omega) = -834\mathrm{Nm}$$





Visualization



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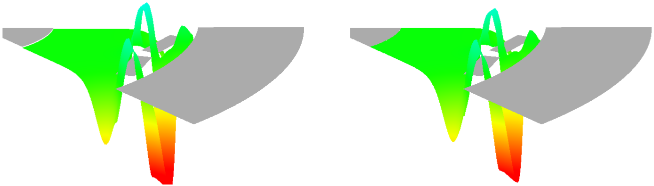

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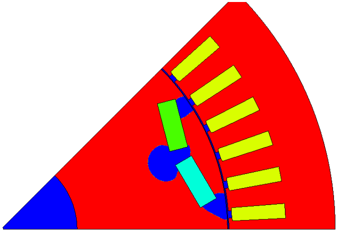
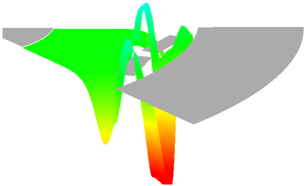
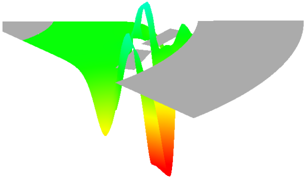


$$\mathcal{J}(\Omega) = -835\text{Nm}$$





Multi-material level set algorithm [Gangl, 2020]

- Tuple of materials $\Omega = (\Omega_f, \Omega_{m_1}, .., \Omega_{m_M}, \Omega_a)$
- Consider all possible material perturbations $\Rightarrow$ Topological derivative in $\mathbb{R}^{N-1}$
- Describe Ω by $N-1$-dim. level set function

$$x \in \Omega_k \Leftrightarrow \psi(x) \in S_k \Leftrightarrow N_k \psi(x) > 0$$

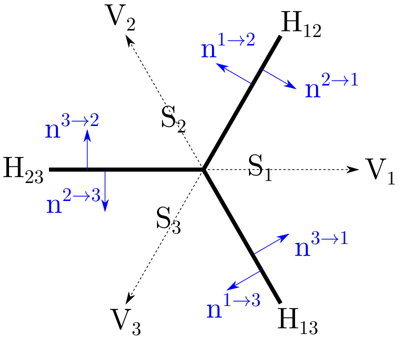
$$N_k := \left((n^{i \to k})^T\right)_{i \neq k}$$

- Define generalized topological derivative

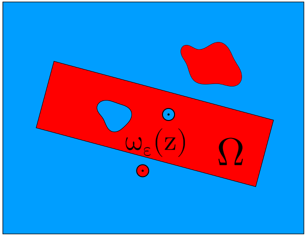
$$\mathrm{d}\mathcal{J}(\Omega)^i(x) = \left(\mathrm{d}\mathcal{J}(\Omega)^{i \to j}(x)\right)^T_{j \neq i},$$

$$g_\Omega(x) := \sum_{i=1}^{N} \chi_{\Omega_i}(x) N_i^{-1} \mathrm{d}\mathcal{J}(\Omega)^i(x)$$

- Optimality condition $\psi = c g_\Omega, c > 0$
- Volume constraint $\sum_{k \in \mathcal{M}_\mathcal{V}} |\Omega_k| \leq V^*$



How to compute the topological derivative [Gangl 2016]



$$\mathrm{d}\mathcal{J}(\Omega)(z) = \lim_{\varepsilon \to 0} \frac{\mathcal{J}(\Omega_\varepsilon(z)) - \mathcal{J}(\Omega)}{|\omega_\varepsilon|}$$

Average torque

$$\frac{1}{N} \sum_{n=0}^{N-1} r_\Gamma \int_\Gamma \nu_0 \mathrm{curl}\, u_\Omega^n \cdot n_\Gamma \mathrm{curl}\, u_\Omega^n \cdot n_\Gamma^\perp \ \mathrm{d}s$$

$$\mathrm{curl}\, h_\Omega(\mathrm{curl}\, u_\Omega^n) = j^n$$

- Compute states u^n and adjoints p^n

$$\begin{aligned} \mathrm{d}\mathcal{J}(\Omega)(z) = \frac{1}{N} \sum_{n=0}^{N-1} \frac{1}{|\omega|} (h_\omega(\mathrm{curl}\,{}_\xi K_{\mathrm{curl}\, u^n(z)} + \mathrm{curl}\, u^n(z)) - h_\omega(\mathrm{curl}\, u^n(z)) \\ - h'_\omega(\mathrm{curl}\, u^n(z))\mathrm{curl}\,{}_\xi K_{\mathrm{curl}\, u^n(z)}, \mathrm{curl}\, p^n(z))_{\mathbb{R}^2} \\ + \frac{1}{|\omega|}((h'_{\Omega^c}(\mathrm{curl}\, u^n(z)) - h'_\Omega(\mathrm{curl}\, u^n(z))\mathrm{curl}\,{}_\xi K_{\mathrm{curl}\, u^n(z)}, \mathrm{curl}\, p^n(z))_\omega \\ + (h_{\Omega^c}(\mathrm{curl}\, u^n(z)) - h_\Omega(\mathrm{curl}\, u^n(z))), \mathrm{curl}\, p^n(z))_2 \end{aligned}$$

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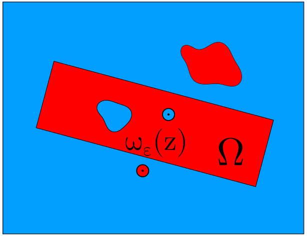
- Formula depends on $\mathrm{curl}\, u^n(z), \mathrm{curl}\, p^n(z) \in \mathbb{R}^2$, timesteps n are independent

$$\mathrm{d}\mathcal{J}(\Omega)(z) = \frac{1}{N} \sum_{n=0}^{N-1} f(\mathrm{curl}\, u^n(z), \mathrm{curl}\, p^n(z))$$

- Take samples of $f : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ in offline phase
- Online evaluation is as cheap as other methods

How to compute the topological derivative [Gangl 2016]



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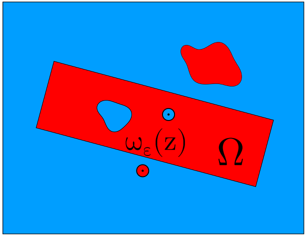
How to compute the topological derivative [Gangl 2016]



$$\mathrm{d}\mathcal{J}(\Omega)(z) = \lim_{\varepsilon\to 0} \frac{\mathcal{J}(\Omega_\varepsilon(z)) - \mathcal{J}(\Omega)}{|\omega_\varepsilon|}$$

Average torque

$$\frac{1}{N}\sum_{n=0}^{N-1} r_\Gamma \int_\Gamma \nu_0 \mathrm{curl}\, u_\Omega^n \cdot n_\Gamma \mathrm{curl}\, u_\Omega^n \cdot n_\Gamma^\perp \ \mathrm{d}s$$

$$\mathrm{curl}\, h_\Omega(\mathrm{curl}\, u_\Omega^n) = j^n$$

- Compute states u^n and adjoints p^n

$$\begin{aligned}\mathrm{d}\mathcal{J}(\Omega)(z) = \frac{1}{N}\sum_{n=0}^{N-1} \frac{1}{|\omega|}\big(&h_\omega(\mathrm{curl}_\xi K_{\mathrm{curl}\, u^n(z)} + \mathrm{curl}\, u^n(z)) - h_\omega(\mathrm{curl}\, u^n(z)) \\ &- h'_\omega(\mathrm{curl}\, u^n(z))\mathrm{curl}_\xi K_{\mathrm{curl}\, u^n(z)}, \mathrm{curl}\, p^n(z)\big)_{\mathbb{R}^2} \\ &+ \frac{1}{|\omega|}((h'_{\Omega^c}(\mathrm{curl}\, u^n(z)) - h'_\Omega(\mathrm{curl}\, u^n(z))\mathrm{curl}_\xi K_{\mathrm{curl}\, u^n(z)}, \mathrm{curl}\, p^n(z))_\omega \\ &+ (h_{\Omega^c}(\mathrm{curl}\, u^n(z)) - h_\Omega(\mathrm{curl}\, u^n(z))), \mathrm{curl}\, p^n(z))_2\end{aligned}$$

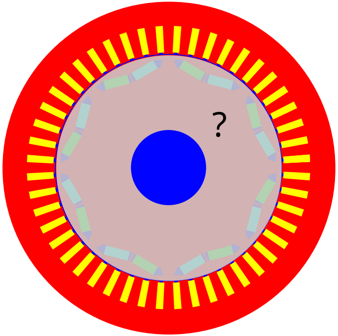
$$(h_\omega(\mathrm{curl}\, K_U + U) - h_\omega(U), \mathrm{curl}\, v)_{\mathbb{R}^2} = -(h_{\Omega^c}(U) - h_\Omega(U), \mathrm{curl}\, v)_\omega, \quad \forall v \in BL$$

- Formula depends on $\mathrm{curl}\, u^n(z), \mathrm{curl}\, p^n(z) \in \mathbb{R}^2$, timesteps n are independent

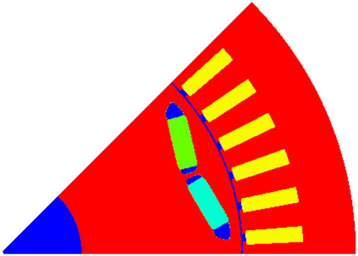
$$\mathrm{d}\mathcal{J}(\Omega)(z) = \frac{1}{N}\sum_{n=0}^{N-1} f(\mathrm{curl}\, u^n(z), \mathrm{curl}\, p^n(z))$$

- Take samples of $f : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ in offline phase
- Online evaluation is as cheap as other methods

Torque maximization

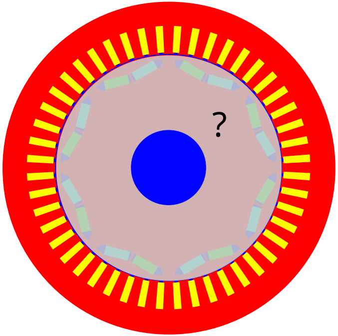


- Single operating point 16000 rpm, 640Nm
- Fixed direction of rotation
- Restrict magnet volume $|M| \leq 10\%|D|$
- Maximize average torque

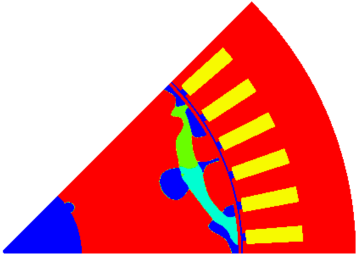


$\overline{T} = 640\mathrm{Nm}$

Torque maximization

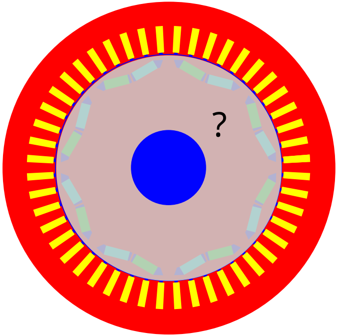


- Single operating point 16000 rpm, 640Nm
- Fixed direction of rotation
- Restrict magnet volume $|M| \leq 10\%|D|$
- Maximize average torque

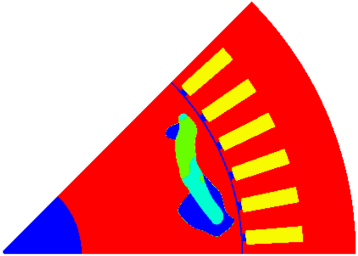


$\overline{T} = 665\mathrm{Nm}$

Torque maximization

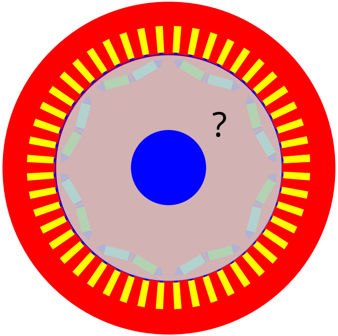


- Single operating point 16000 rpm, 640Nm
- Fixed direction of rotation
- Restrict magnet volume $|M| \leq 10\%|D|$
- Maximize average torque

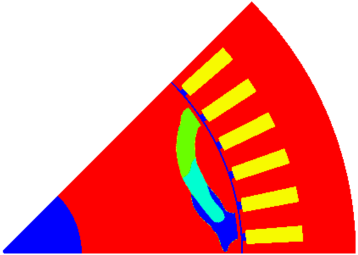


$\overline{T} = 726\mathrm{Nm}$

Torque maximization

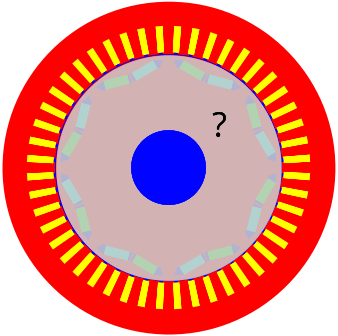


- Single operating point 16000 rpm, 640Nm
- Fixed direction of rotation
- Restrict magnet volume $|M| \leq 10\%|D|$
- Maximize average torque

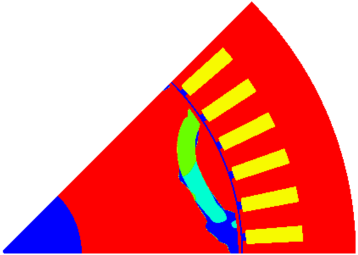


$\overline{T} = 828\mathrm{Nm}$

Torque maximization

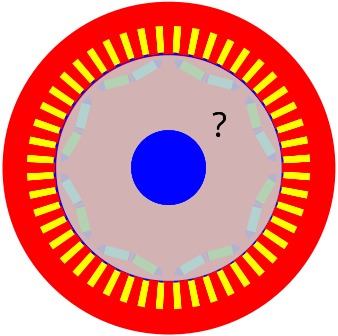


- Single operating point 16000 rpm, 640Nm
- Fixed direction of rotation
- Restrict magnet volume $|M| \leq 10\%|D|$
- Maximize average torque

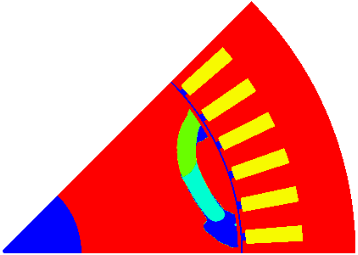


$\overline{T} = 847\mathrm{Nm}$

Torque maximization

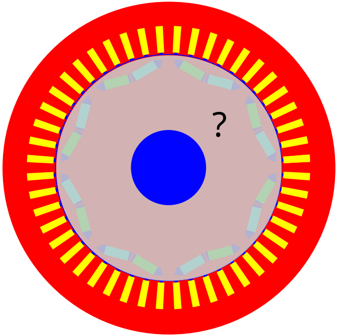


- Single operating point 16000 rpm, 640Nm
- Fixed direction of rotation
- Restrict magnet volume $|M| \leq 10\% |D|$
- Maximize average torque

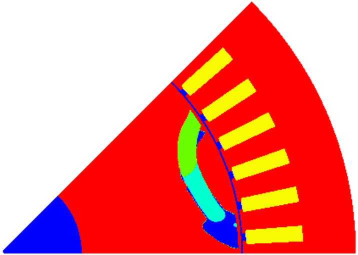


$\overline{T} = 857\mathrm{Nm}$

Torque maximization

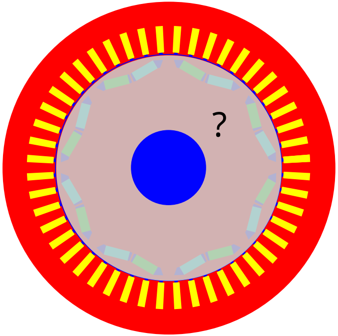


- Single operating point 16000 rpm, 640Nm
- Fixed direction of rotation
- Restrict magnet volume $|M| \leq 10\%|D|$
- Maximize average torque

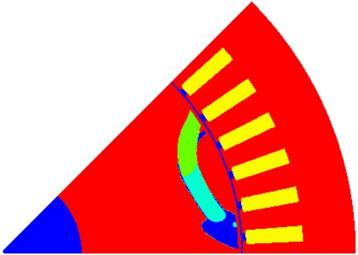


$\overline{T} = 859\mathrm{Nm}$

Torque maximization

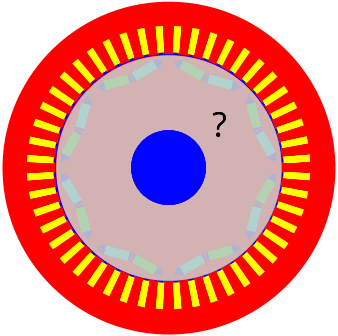


- Single operating point 16000 rpm, 640Nm
- Fixed direction of rotation
- Restrict magnet volume $|M| \leq 10\%|D|$
- Maximize average torque

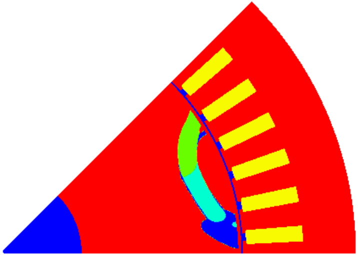


$\overline{T} = 861\mathrm{Nm}$

Torque maximization

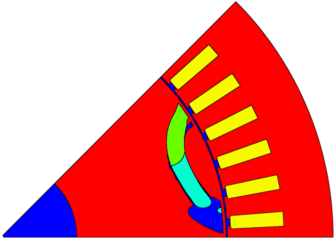


- Single operating point 16000 rpm, 640Nm
- Fixed direction of rotation
- Restrict magnet volume $|M| \leq 10\% |D|$
- Maximize average torque



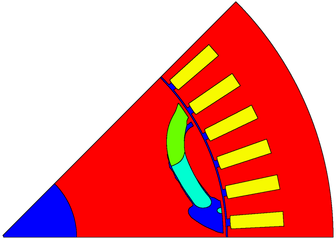
$\overline{T} = 862\mathrm{Nm}$

Nice torque, but...

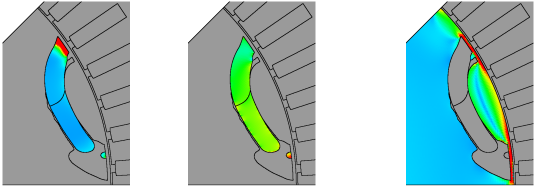


$\overline{T} = 862\mathrm{Nm}$

Nice torque, but...

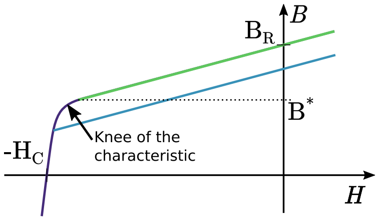


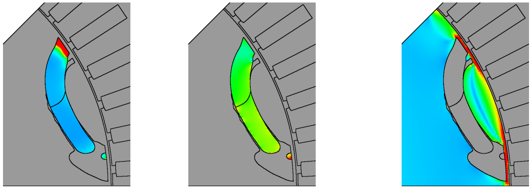
$\overline{T} = 862\mathrm{Nm}$



$\vartheta_{\max} = 210°$ $\quad B_{\min} = -0.3\mathrm{T}$ $\quad \sigma^{vM}_{\max} = 1974\mathrm{MPa}$

Nice torque, but...

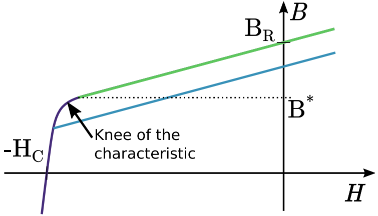


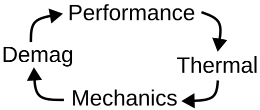


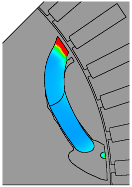
$\vartheta_{\max} = 210°$ $B_{\min} = -0.3\mathrm{T}$ $\sigma^{vM}_{\max} = 1974\mathrm{MPa}$

$\vartheta^* = 90°$ $B^* = 0.56\mathrm{T}$ $S^* = 500\mathrm{MPa}$

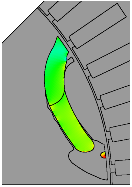
Nice torque, but...



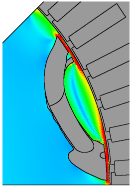




$\vartheta_{\max} = 210°$
$\vartheta^* = 90°$

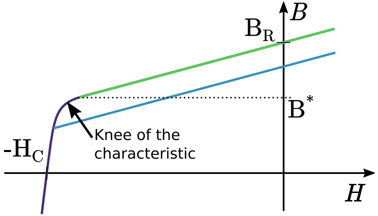


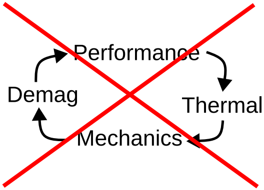
$B_{\min} = -0.3\mathrm{T}$
$B^* = 0.56\mathrm{T}$

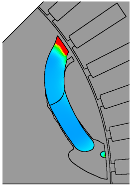


$\sigma^{vM}_{\max} = 1974\mathrm{MPa}$
$S^* = 500\mathrm{MPa}$

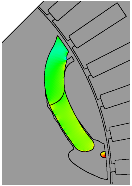
Nice torque, but...



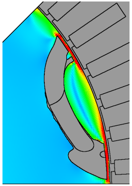




$\vartheta_{\max} = 210°$
$\vartheta^* = 90°$



$B_{\min} = -0.3\mathrm{T}$
$B^* = 0.56\mathrm{T}$



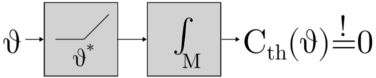
$\sigma^{vM}_{\max} = 1974\mathrm{MPa}$
$S^* = 500\mathrm{MPa}$

$\rightarrow$ All-in-one, add constraints to optimization

Constraints

- Temperature

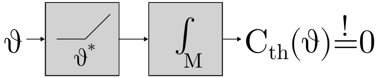
$$-\mathrm{div}\lambda\nabla\vartheta = \frac{\sigma_M}{T}\int_0^T (\partial_t a - \overline{\partial_t a})^2 \, \mathrm{d}t$$



Constraints

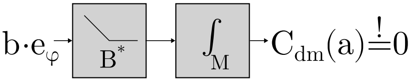
- Temperature

$$-\mathrm{div}\lambda\nabla\vartheta = \frac{\sigma_M}{T}\int_0^T (\partial_t a - \overline{\partial_t a})^2 \,\mathrm{d}t$$



- Magnetic fields

$$\mathrm{curl}\, h(\mathrm{curl}\, a) = j$$



Constraints

- Temperature

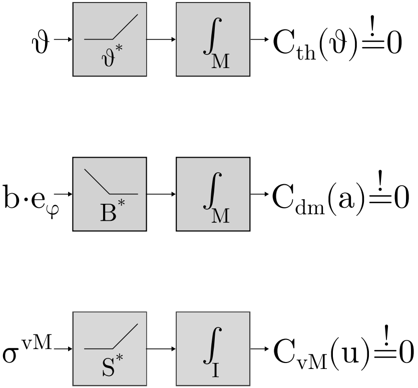
$$-\mathrm{div}\lambda\nabla\vartheta = \frac{\sigma_M}{T}\int_0^T (\partial_t a - \overline{\partial_t a})^2 \,\mathrm{d}t$$

- Magnetic fields

$$\mathrm{curl}\, h(\mathrm{curl}\, a) = j$$

- Mechanical stresses

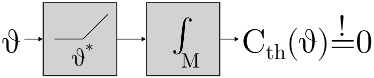
$$-\mathrm{div}\sigma(u) = \rho_I\omega^2 x, \quad \sigma^{\mathrm{vM}} = \sqrt{\tfrac{1}{2}\mathbb{B}\sigma : \sigma}$$



Constraints

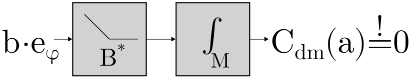
- Temperature

$$-\mathrm{div}\lambda\nabla\vartheta = \frac{\sigma_M}{T}\int_0^T (\partial_t a - \overline{\partial_t a})^2 \,\mathrm{d}t$$



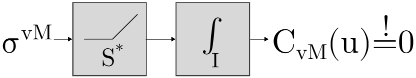
- Magnetic fields

$$\mathrm{curl}\, h(\mathrm{curl}\, a) = j$$



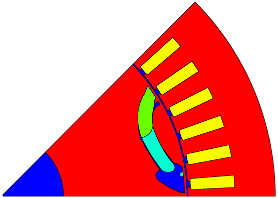
- Mechanical stresses

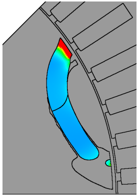
$$-\mathrm{div}\sigma(u) = \rho_I \omega^2 x, \quad \sigma^{\mathrm{vM}} = \sqrt{\tfrac{1}{2}\mathbb{B}\sigma : \sigma}$$



$$\rightarrow \min_{\Omega} -\overline{T}(a(\Omega)) + w_{\mathrm{th}}\, C_{\mathrm{th}}(\vartheta(\Omega)) + w_{\mathrm{dm}}\, C_{\mathrm{dm}}(a^{\mathrm{d}}(\Omega)) + w_{\mathrm{vM}}\, C_{\mathrm{vM}}(u(\Omega))$$

Temperature constraint





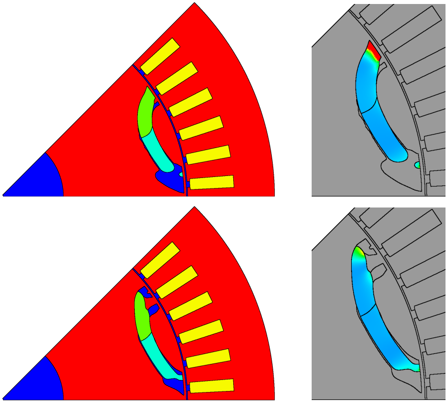
$$\vartheta^* = 90^\circ,$$

$$\min_{\Omega, |\Omega_m| \le V^*} \mathcal{T}(\Omega) + w_{\mathrm{th}} \mathcal{C}_{\mathrm{th}}(\Omega)$$

Constr.	$\mathcal{T}(\Omega)$	$\max \vartheta$
-	862	210

P. Gangl, N. Krenn, "Multi-material topology optimization of electric machines under maximum temperature and stress constraints", 2025

Temperature constraint



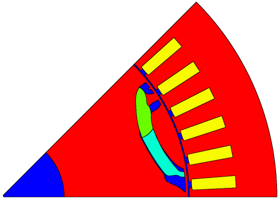
$$\vartheta^* = 90°,$$

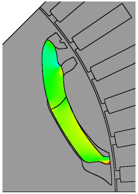
$$\min_{\Omega, |\Omega_m| \leq V^*} \mathcal{T}(\Omega) + w_{\text{th}} \mathcal{C}_{\text{th}}(\Omega)$$

Constr.	$\mathcal{T}(\Omega)$	$\max \vartheta$
-	862	210
th	845	92

P. Gangl, N. Krenn, "Multi-material topology optimization of electric machines under maximum temperature and stress constraints", 2025

Demag constraint





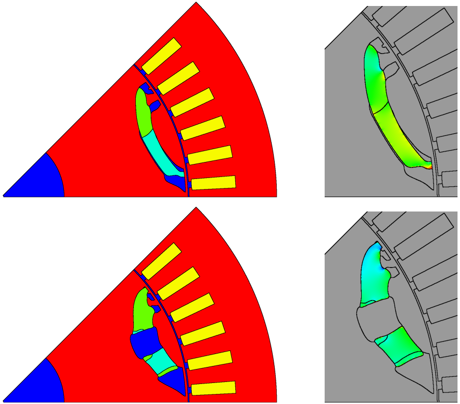
$B^* = 0.56\mathrm{T}$

$$\min_{\Omega, |\Omega_m| \leq V^*} \mathcal{T}(\Omega) + w_{\mathrm{th}} \mathcal{C}_{\mathrm{th}}(\Omega) + w_{\mathrm{dm}} \mathcal{C}_{\mathrm{dm}}(\Omega)$$

Constr.	$\mathcal{T}(\Omega)$	$\max \vartheta$	$B_{\min}$
-	862	210	-0.3
th	845	92	-0.32

P. Gangl, N. Krenn, "Multi-material topology optimization of an electric machine considering demagnetization", 2024

Demag constraint



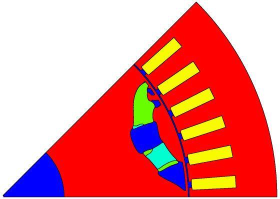
$$B^* = 0.56\mathrm{T}$$

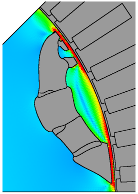
$$\min_{\Omega, |\Omega_m| \leq V^*} \mathcal{T}(\Omega) + w_{\mathrm{th}}\mathcal{C}_{\mathrm{th}}(\Omega) + w_{\mathrm{dm}}\mathcal{C}_{\mathrm{dm}}(\Omega)$$

Constr.	$\mathcal{T}(\Omega)$	max ϑ	$B_{\min}$
-	862	210	-0.3
th	845	92	-0.32
th,dm	774	92	0.41

P. Gangl, N. Krenn, "Multi-material topology optimization of an electric machine considering demagnetization", 2024

Stress constraint





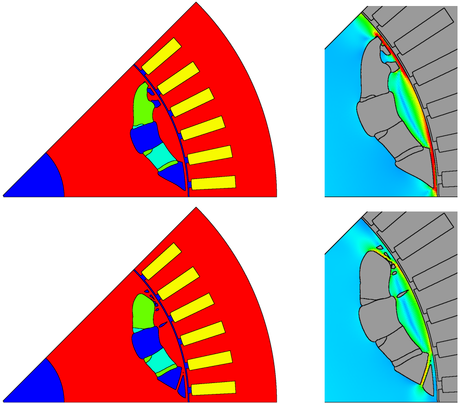
$$S^* = 500\text{MPa}$$

$$\min_{\Omega, |\Omega_m| \le V^*} \mathcal{T}(\Omega) + \ldots + w_{\text{vM}} \mathcal{C}_{\text{vM}}(\Omega)$$

Constr.	$\mathcal{T}(\Omega)$	$\max \vartheta$	$B_{\min}$	$\max \sigma^{\text{vM}}$
-	862	210	-0.3	1974
th	845	92	-0.32	2007
th,dm	774	92	0.41	2338

P. Gangl, N. Krenn, "Multi-material topology optimization of electric machines under maximum temperature and stress constraints", 2025

Stress constraint

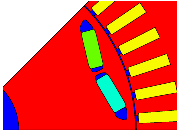


$$S^* = 500\text{MPa}$$

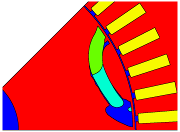
$$\min_{\Omega, |\Omega_m| \leq V^*} \mathcal{T}(\Omega) + \ldots + w_{\text{vM}} \mathcal{C}_{\text{vM}}(\Omega)$$

Constr.	$\mathcal{T}(\Omega)$	$\max \vartheta$	$B_{\min}$	$\max \sigma^{\text{vM}}$
-	862	210	-0.3	1974
th	845	92	-0.32	2007
th,dm	774	92	0.41	2338
th,dm,vM	736	88	0.39	498

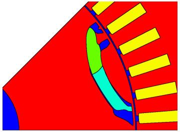
P. Gangl, N. Krenn, "Multi-material topology optimization of electric machines under maximum temperature and stress constraints", 2025



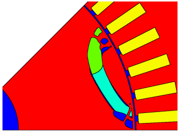
Reference design



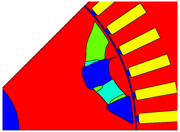
Torque maximization



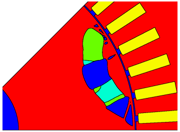
Temp. constraint



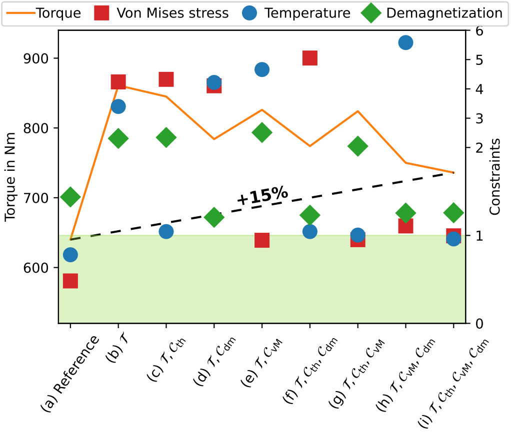
Stress constraint



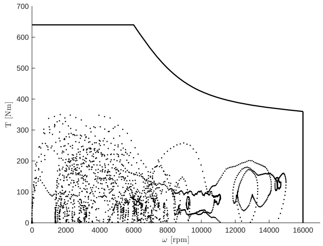
Demag constraint



All constraints



Drive cycle



- Replace average torque by efficiency

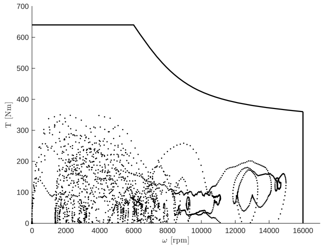
$$\mathcal{E}(\Omega) = \frac{E^m}{E^e} = \frac{\sum t_k P_k^m}{\sum t_k (P_k^m + R_S I_k^2(\Omega)/2)}$$

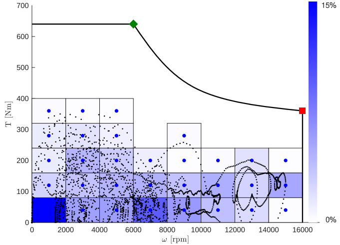
- Current given via MTPA problem

$$I_k(\Omega) = \min_{I,\beta} I, \text{ subject to } \overline{T}(\Omega, I, \beta) = T_k$$

N. Krenn et. al, "Electro-thermal topology optimization of an electric machine by the topological derivative considering drive cycles", IEEE, 2025

Drive cycle





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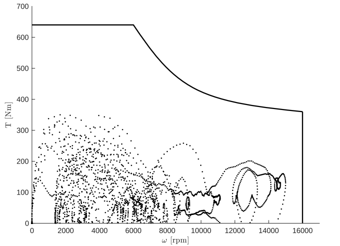

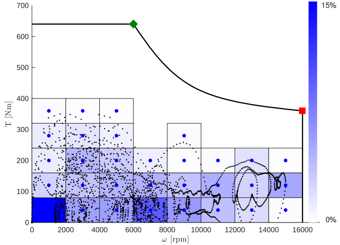

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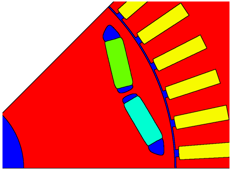
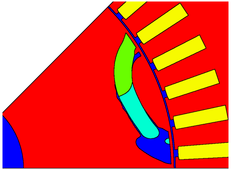
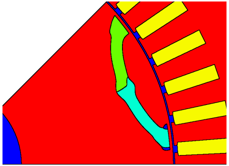


$\mathcal{E}(\Omega) = 99.610\%$



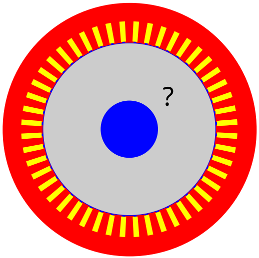
$\mathcal{E}(\Omega) = 99.737\%$

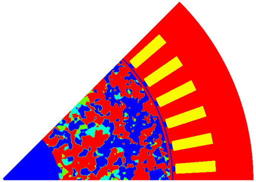


$\mathcal{E}(\Omega) = 99.778\%$

N. Krenn et. al, "Electro-thermal topology optimization of an electric machine by the topological derivative considering drive cycles", IEEE, 2025

But what about...





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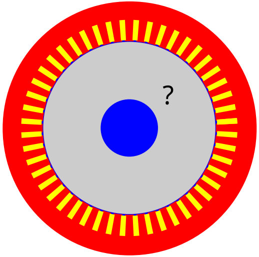
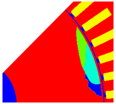
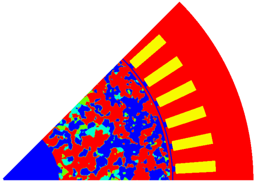



859



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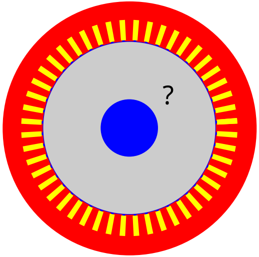
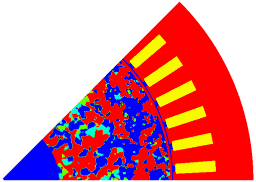
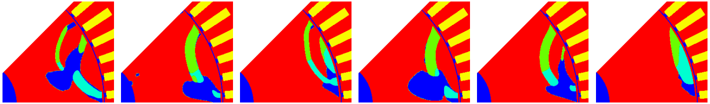
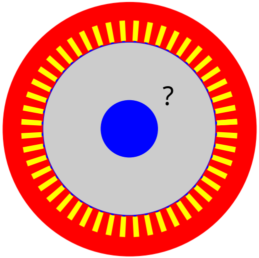


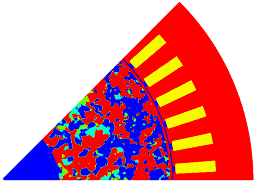


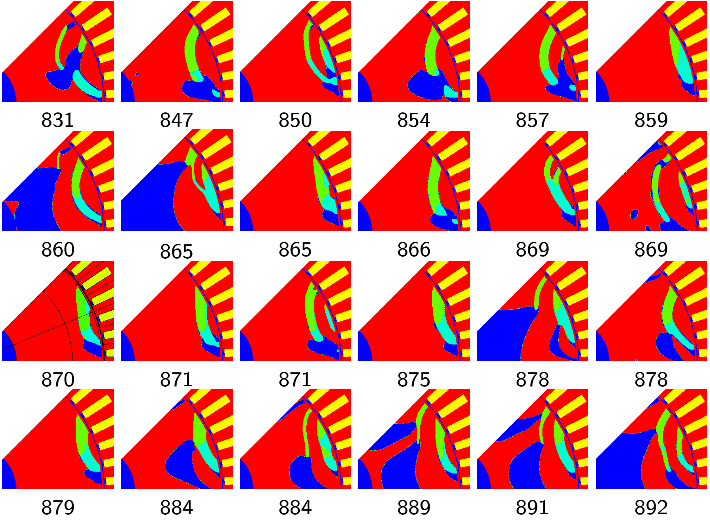


831 847 850 854 857 859

But what about...







Outlook

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  - L. Beck et. al, "Computing Multiple Local Minimizers for the Topology Optimization of Bipolar Plates in Electrolysis Cells", arxiv, 2024
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- Implement active short circuit
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Thank you for your attention!

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